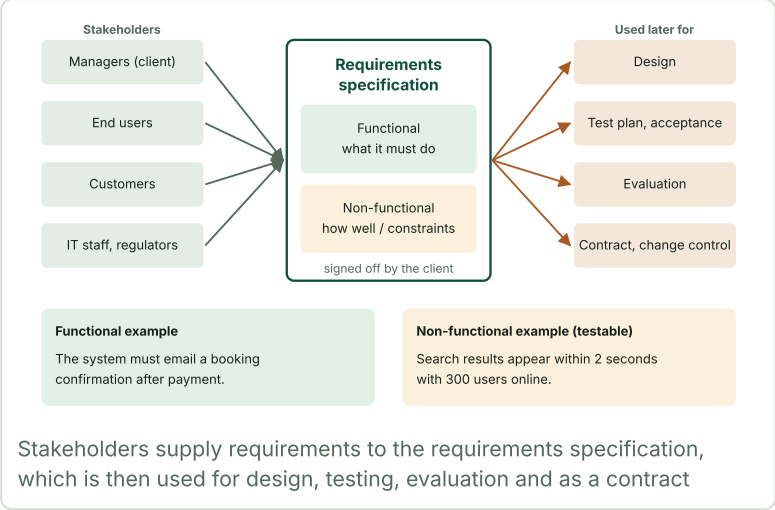
Stakeholders, requirements and the system specification

AP04 FACT FILE

Specification: students should be able to

- Identify the stakeholders in a systems development project and describe their differing needs
- Distinguish between functional and non-functional user requirements
- Identify functional and non-functional requirements for systems such as stock control, reservation, payroll and billing
- Describe the purpose and contents of the system (requirements) specification produced at the end of analysis
- Explain how the role of the specification has changed with the move towards more flexible, Agile approaches

Key words: stakeholder, client, end user, user requirements, functional requirement, non-functional requirement, quality attribute, constraint, system specification, requirements specification, performance, reliability, availability, security, usability, testable, MoSCoW, Agile



Stakeholders supply requirements to the requirements specification, which is then used for design, testing, evaluation and as a contract

Requirements: the bridge from analysis to design

Once fact-finding is complete, the systems analyst turns the evidence into a clear statement of user requirements: everything the new system must do and the standards it must meet. These requirements are agreed with the client and recorded in the system specification, which becomes the reference point for the rest of the project. Designers use it to plan the system, programmers build to it, testers check against it and the final evaluation measures the finished product against it. Requirements are divided into two kinds. Functional requirements state what the system must do, for example recording a booking, calculating a payslip or printing an invoice. Non-functional requirements state the conditions under which it must do those things, such as how quickly, how reliably, how securely and for how many users at once. A useful test is to ask whether the statement describes an action the system carries out or a quality by which that action is judged. Requirements must also reflect the needs of the stakeholders, and different stakeholders frequently want different things, so the analyst has to identify each group's needs, resolve conflicts with the client and write each requirement so that it can later be tested.

Stakeholders

A stakeholder is any person or group who has an interest in a system or is affected by it. Internal stakeholders include the client, usually senior managers who commission the system and pay for it; the end users who operate it every day; and the IT staff who will install, support and maintain it. External stakeholders include customers, suppliers, shareholders and regulators. Regulators matter because systems handling personal data must comply with the UK GDPR and the Data Protection Act 2018, overseen by the Information Commission (formerly the Information Commissioner's Office, ICO), while a payroll system must meet HMRC rules. Stakeholders' priorities often conflict. A manager might want detailed reports and tight security, while front-line users want a quick, simple screen with as few log-ins as possible, and customers want convenience and privacy. The analyst must identify these groups during fact-finding, record their requirements, and help the client decide priorities. Involving end users early has a further benefit: people who feel consulted are more likely to accept the new system, while staff who feel it has been imposed on them may resist it or use it badly, which can undermine even a technically excellent solution.

Functional requirements

A functional requirement describes something the system must do or allow its users to do. Functional requirements cover inputs, processing, outputs and stored data, and are usually written in a form such as 'The system must allow authorised staff to add and edit customer records' or 'The system must produce a monthly report of unpaid accounts'. For a stock control system, typical functional requirements are keeping track of current stock levels including items already on order, recording when each batch arrived so that stock can be rotated on a first-in, first-out basis, alerting staff when an item falls to its reorder level and producing management reports. For a payroll system they include adding and removing employees, entering hours worked, changing pay rates, calculating gross pay, deductions and net pay, and producing payslips. A billing system must calculate charges from usage and tariffs, add VAT, issue bills and record payments. A reservation system must show availability, take bookings and payments, and send confirmations. Note that backing up data can itself be a functional requirement, because the system must provide a backup facility, while how often backups happen is a non-functional condition.

Non-functional requirements

Non-functional requirements describe the conditions that the system's behaviour must satisfy rather than the behaviour itself. They are sometimes called quality attributes because they provide the criteria by which the system is judged. The main categories are performance, including response time, throughput and use of processor and memory; availability, meaning the hours or percentage of time the system can be used; reliability, meaning how rarely it fails and how accurately it works; security, covering who has access, who has priority, password rules and the frequency of backups; capacity, such as the number of simultaneous users; usability and accessibility; maintainability; and legal compliance. For example, a requirement that changes to shared data must be visible to every user within two seconds is non-functional, because it does not add a new function but sets a standard for an existing one. Good non-functional requirements are measurable. 'The website should be fast' cannot be tested, whereas 'search results must appear within two seconds with 500 users logged in' can. Weak answers in assessments often state a vague quality without any measure or link to the scenario.

Requirements depend on perspective

The same system can be described quite differently depending on whose requirements are being considered. Take an online ferry booking system. From the ferry company's point of view, functional requirements might include presenting sailings that match the customer's choices, monitoring how full each sailing is, raising fares automatically as capacity is approached and keeping a booking history for registered members. Its non-functional requirements would include restricting management screens to authorised staff, being available at all times, being backed up daily and coping with many simultaneous users without selling the same space twice. From the customer's point of view, the functional requirements are things like searching for sailings, choosing a cabin, paying by card, receiving an email confirmation and amending a booking online, while non-functional requirements include fast confirmation, keeping card details secure and working well on a mobile phone. When a question specifies a perspective, answers must match it: listing what the customer wants when the question asks for the organisation's requirements will not earn marks. Thinking about each stakeholder in turn is also a useful way to generate a complete set of requirements during analysis.

The system specification

The system specification, also called the requirements specification, is the main document produced at the end of analysis. It states precisely what the client expects the software to do and sets out a schedule for producing it. Typical contents are the scope and objectives of the project, a description of the current system and its problems, the stakeholders, the functional and non-functional requirements, the data requirements, data flow diagrams, and constraints such as budget, deadline and hardware. Historically, the specification formed the basis of a legally binding contract between the organisation buying the system and the supplier. The agreement worked both ways: the client could not later demand extra or different features, while the supplier was obliged to deliver everything specified, and changes led to penalties or revised charges. This suited an era with few software suppliers, who could plan their workforce around fixed deliverables. Today there are many more suppliers and development skills are widespread, and Agile approaches deliver software in short iterations with frequent client feedback. As a result, modern specifications tend to be less prescriptive and include agreed processes for changing requirements, while still giving both sides a clear shared understanding.

Key terms from the mark schemes

Stakeholder
A person or group with an interest in a system or who is affected by it, such as managers, end users, customers and regulators.

End user
A person who operates the system directly in their day-to-day work or as a customer.
Credited: User

Functional requirement
A statement of something the system must do, such as an input, process, calculation or output.
Credited: What the system must do

Non-functional requirement
A condition or constraint on how the system behaves, such as response time, availability, reliability, security or usability, rather than a function it performs.
Credited: Additional conditions the system should meet

Quality attribute
A measurable characteristic used to judge a system, such as performance, reliability or security; non-functional requirements specify these.

System specification
The document produced at the end of analysis that states precisely what the client expects the system to do, including requirements, constraints and schedule, and is agreed by the client.
Credited: Requirements specification

MoSCoW
A method of prioritising requirements as Must have, Should have, Could have and Won't have this time.

Constraint
A limit within which the system must be developed or operate, such as budget, deadline, hardware or legislation.

Testable requirement
A requirement stated precisely enough, usually with a measure, that it can be shown to have been met or not.
Credited: Measurable requirement

Exam focus

Expect a two-mark question asking you to distinguish the two kinds of requirement: one mark for explaining that a functional requirement describes an action the system has to carry out, and one for saying a non-functional requirement is a condition, constraint or quality the system must meet, such as performance or security. Scenario questions then ask for two or three examples of each for a named system such as payroll, stock control, billing or reservations. Each example must clearly belong to that system, so 'add and edit employee details' suits payroll, whereas 'be secure' on its own is too vague; 'only payroll staff can view salary details' is much better. Begin functional requirements with 'The system must allow...' or 'The system must produce...', and give non-functional requirements a category and, where possible, a measure. Read carefully whether the question asks for the organisation's or the customer's perspective. If asked to name the document produced at the end of analysis, the answer is the system (requirements) specification. When explaining its importance, mention that it records what was agreed, guides design, provides criteria for testing and evaluation, and helps avoid disputes. Avoid giving hardware items as requirements.

? Check yourself

1. Define the term stakeholder and give two examples of stakeholders in a new school management information system. [3]
2. A requirements list for a Strabane veterinary practice includes 'Record each animal's vaccinations' and 'Show any pet's record within two seconds'. Using these as examples, explain how a functional requirement differs from a non-functional requirement. [2]
3. A hardware merchant in Newry is replacing its paper-based stock control system. State two functional and two non-functional requirements of the new system. [4]

Answers: 1. Definition: a person or group with an interest in, or affected by, the system (1). Any two examples (1 for both): teachers / pupils / parents / senior leadership / office staff / IT technician / Education Authority / Department of Education (regulator). 2. A functional requirement describes an action the system must carry out / a function, input, process or output, e.g. recording vaccinations (1) / a non-functional requirement describes a condition, constraint or quality the system must meet, such as performance, e.g. displaying a record within two seconds (1) 3. Functional, any two (1 each): record/track current stock levels including items on order / alert staff when stock reaches reorder level / record goods received and sold / produce stock reports / generate purchase orders / support FIFO stock rotation. Non-functional, any two (1 each): stock levels must be accurate/up to date / available during trading hours / accessible only to authorised staff / backed up daily / support multiple simultaneous users / respond within a stated time